Impact of 6G Network Performance on Manufacturing Efficiency in Smart Factories

A Descriptive and Diagnostic Analysis of Network Latency, Packet Loss, and Manufacturing Outcomes in a Smart-Factory Dataset

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Research Paper

Submitted by

Isha Kalra

Bachelor of Computer Applications (BCA) — 3rd Year

Dewan V.S. Group of Institutions, Meerut

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ABSTRACT

Smart manufacturing environments depend on communication networks to coordinate machines, transmit sensor data, and support monitoring and operational decisions, which makes network latency and packet loss natural candidates for monitoring. This study examines whether network-performance measurements are associated with manufacturing outcomes in a smart-factory dataset of 100,000 observations from 50 machines, recorded between 1 January and 10 March 2025. Using descriptive and diagnostic analytics — a dataset-relative Network Stability Index (NSI), tertile-based network-quality groups, decile comparisons, Spearman rank correlation, a linear latency slope, operation-mode comparisons, and grouping by comparable temperature, vibration, and power-consumption profiles — the analysis relates latency and packet loss to efficiency status, production speed, error rate, and defect rate.

Efficiency status was nearly identical across network-quality groups (Low efficiency 77.75%–77.89%). The Spearman correlation between latency and production speed was −0.0011, the fitted slope was −0.00967 units/hr per millisecond, and no sustained latency sensitivity zone emerged. Packet loss showed a Spearman correlation of −0.0024 with error rate, and mean defect rates stayed between 4.95% and 5.07% across packet-loss deciles. The Packet Loss Impact Ratio was 1.54%, an extreme-decile contrast that does not reflect a consistent progressive decline. Results were similar across operation modes and within comparable (not identical) mechanical-condition profiles.

The findings do not establish network performance as an independent driver of manufacturing efficiency in this dataset, and no validated latency-tolerance benchmark or packet-loss risk zone could be derived. The study evaluates the supplied smart-factory dataset only: its provenance is undocumented, several statistical regularities suggest possible simulation, and the analysis does **not** independently validate a real 6G deployment. The results support monitoring network indicators alongside machine and process conditions, and validating any threshold with richer, synchronized data before operational use. The analysis is observational and does not demonstrate causation.

Keywords: Smart Manufacturing; Network Latency; Packet Loss; Network Stability Index; Manufacturing Efficiency; Operation Modes; Industrial Analytics

1. INTRODUCTION

1.1 Background

A smart factory connects machines, sensors, industrial controllers, and monitoring systems so that information about production can be collected and acted upon. Industrial communication is changing as Internet-of-Things and cyber-physical-system concepts enter automation (Wollschlaeger et al., 2017), and the industrial Internet-of-Things literature lists real-time performance among its central challenges, alongside energy efficiency, coexistence, interoperability, and security (Sisinni et al., 2018). Manufacturers are also collecting growing volumes of operational data to support data-driven decisions (Tao et al., 2018). In this setting, two basic indicators describe the quality of a communication link: *latency*, the delay with which data travels between devices (measured here in milliseconds, ms), and *packet loss*, the percentage of transmitted data units that fail to arrive.

Standards work reflects the same concern. The 3GPP service requirements for cyber-physical control applications state that supporting communication services must be ultra-reliable and, in some cases, offer very low end-to-end latency (3GPP, TS 22.104, Release 16 and later). Forward-looking visions of 6G and of IMT-2030 describe enhanced communication capabilities intended to support demanding applications, including autonomous systems (Saad et al., 2020; ITU-R, 2023). Together, these sources explain why network measurements are routinely proposed as monitoring variables in connected manufacturing.

These are, however, general motivations. They do not show that, in a particular factory, a given change in latency or packet loss will change production speed, errors, or defects. Manufacturing outcomes also vary with machine condition, operating mode, workload, temperature, vibration, and power consumption. This paper therefore keeps four layers distinct: (i) the general motivation for studying network performance; (ii) what established literature says about industrial communication and control; (iii) what this particular dataset shows; and (iv) what can reasonably be recommended on that basis.

1.2 Problem Statement

Although network performance is an important consideration in connected manufacturing environments, the presence of latency or packet loss does not by itself demonstrate that efficiency or product quality has deteriorated because of those conditions. A data-driven investigation is needed to examine which relationships are actually visible in the available observations, how large they are, and which conclusions the evidence supports. This project asks whether network-performance measurements are associated with manufacturing efficiency, production speed, operational errors, and quality defects; whether any latency or packet-loss range is associated with a sustained deterioration; and whether the picture changes across operation modes or when mechanical conditions are held comparable.

1.3 Research Objectives

The overall objective is to analyze the relationship between network performance and manufacturing outcomes in a smart-factory dataset, focusing on latency, packet loss, network stability, efficiency status, production speed, operational errors, quality defects, and operation modes, and to derive evidence-based recommendations for network monitoring and manufacturing operations. Specifically, the study aims to:

1. Characterize the observed distributions and temporal patterns of latency and packet loss.
2. Examine how efficiency status varies across the defined network-quality groups.
3. Investigate the relationships between latency and production speed, and between packet loss and error rate and defect rate.
4. Compare network-related patterns across the Active, Idle, and Maintenance operation modes.
5. Examine network-related outcomes within comparable mechanical-condition groups.
6. Translate the results into evidence-based monitoring recommendations.

1.4 Research Questions

- **RQ1 (Network performance):** What are the observed distributions and temporal patterns of network latency and packet loss?
- **RQ2 (Network quality and efficiency):** How does the distribution of efficiency status vary across the defined network-quality groups?
- **RQ3 (Latency and production):** What relationship, if any, is observed between network latency and production speed?
- **RQ4 (Packet loss and errors):** What relationship, if any, is observed between packet loss and error rate?
- **RQ5 (Packet loss and quality):** How do observed defect rates vary across packet-loss conditions?
- **RQ6 (Operation modes):** Do the observed relationships differ across Active, Idle, and Maintenance modes?
- **RQ7 (Comparable mechanical conditions):** Do network-related differences in outcomes remain apparent when observations are grouped by similar mechanical conditions?
- **RQ8 (Operational recommendations):** What can the observed patterns reasonably suggest about network monitoring, and what additional evidence is required before production-grade thresholds are defined?

1.5 Scope and Contributions

The study uses the supplied manufacturing dataset and applies descriptive and diagnostic analytics. It defines and examines selected network-performance indicators, investigates operational and quality relationships, and produces monitoring recommendations. An interactive Streamlit dashboard supports exploration of the same results, but this paper is written to be read independently of it. The study does **not** develop a new 6G protocol, validate a live industrial deployment, train a predictive model, or establish causal effects. Its contribution is a transparent, reproducible analysis in which network indicators, efficiency, speed, errors, defects, operation modes, and mechanical conditions are examined within one framework, with null and weak findings, data-structure caveats, and the limits of inference reported explicitly.

2. LITERATURE REVIEW

The review is deliberately selective: it covers only sources that bear on the research questions, and it separates what each source establishes from what the present dataset can and cannot show. Table 1 summarizes the sources and their relevance.

2.1 Industrial Communication in Smart Manufacturing

Wollschlaeger et al. (2017) review how Internet-of-Things and cyber-physical-system concepts are reshaping industrial automation, covering Ethernet time-sensitive networking and the role of fifth-generation networks in automation. Sisinni et al. (2018) clarify the relationship between IoT, industrial IoT, and Industry 4.0 and identify real-time performance, coexistence, interoperability, energy efficiency, and security as the principal challenges of industrial deployment. Tao et al. (2018) discuss the role of big data in smart manufacturing and propose a framework in which manufacturing data support data-driven strategies. The three sources agree on the direction of travel — more connected equipment and more data — but none of them supplies a general rule linking a measured network value to a manufacturing outcome.

2.2 Network Performance and Manufacturing Operations

The 3GPP specification for cyber-physical control applications describes communication services that must be ultra-reliable and, in some cases, provide very low end-to-end latency, with factories of the future a principal use case (3GPP, TS 22.104). The 6G vision of Saad et al. (2020) notes that, at the time of writing, the architectural and performance components of 6G were still largely undefined, and ITU-R Recommendation M.2160-0 sets out the framework, usage scenarios, and capabilities for IMT-2030 (ITU-R, 2023). These sources describe *service targets and design intentions*. They do not report outcomes measured in a specific production line, and the present dataset does not state which network technology generated its measurements.

2.3 Network Latency and Packet Loss

The networked-control literature provides the clearest theoretical link between network behavior and operational performance. Zhang et al. (2001) examine how sampling, network-induced delay, and packet dropout affect the stability of feedback control over a network, and Hespanha et al. (2007) survey results on networked control systems in which communication between sensors, actuators, and controllers is subject to limited packet rates, delay, and loss. Schenato et al. (2007) model unreliable networks stochastically, assigning probabilities to successful packet delivery, and characterize how network reliability affects feedback-loop performance; their results also depend on whether delivery is acknowledged. The common message is that the consequences of delay and loss depend on the control design, the sampling arrangement, the loss probability, and the protocol. They are results about *closed-loop control with explicit models*, not general statements that a given percentage of packet loss changes factory output.

2.4 Network Reliability and Product Quality

The sources reviewed connect communication quality to control performance and service reliability. Within them, no general relationship translating a packet-loss percentage into a product-defect rate was identified; such a link, if present, would depend on whether the affected traffic participates in a quality-relevant control or inspection loop. This is a statement about the sources reviewed here, not a claim that no such evidence exists. It does mean that a weak packet-loss–defect association in a given dataset is not necessarily at odds with the literature, and that a strong one could not be taken as general.

2.5 Research Gap and Project Motivation

This review makes no claim that the topic is unstudied. The narrower motivation is practical: the project applies one consistent descriptive and diagnostic analysis to a single dataset in which network measurements appear alongside efficiency status, production speed, errors, defects, operation modes, and mechanical conditions, and it reports clearly what that dataset does and does not show. Where the literature establishes mechanisms under controlled or modeled conditions, this analysis asks the complementary, empirical question of whether comparable patterns are visible in the supplied observations.

Table 1. Summary of Reviewed Sources and Their Relevance to This Study

Source	Topic	Focus	Relevance to this study
Wollschlaeger et al. (2017)	Industrial communication	IoT, cyber-physical systems, time-sensitive networking, role of 5G in automation	Context for connected factories
Sisinni et al. (2018)	Industrial IoT	Challenges incl. real-time performance, coexistence, security	Motivates monitoring of real-time performance
Tao et al. (2018)	Data-driven manufacturing	Role of big data and a data-driven smart-manufacturing framework	Rationale for analytics on operational data
3GPP TS 22.104	Service requirements	Ultra-reliable, in some cases very low-latency services for cyber-physical control	Source of general latency/reliability motivation
Zhang et al. (2001)	Networked control	Stability of control over a network (delay, sampling, loss)	Mechanism linking delay/loss to control
Hespanha et al. (2007)	Networked control	Survey of results under rate, delay, and loss limits	Mechanisms depend on design and sampling
Schenato et al. (2007)	Lossy networks	Control and estimation with probabilistic packet loss	Packet-loss effects are model- and protocol-dependent
Saad et al. (2020)	6G vision	Applications, trends, technologies, open problems	Context for the 6G framing
ITU-R M.2160-0 (2023)	IMT-2030 framework	Framework, usage scenarios, capabilities	Context for the 6G framing

Sources are described only to the extent verified from their abstracts or official descriptions; none reports results for this dataset.

3. DATASET AND METHODOLOGY

3.1 Dataset Description

The dataset (file *Thales_Group_Manufacturing.csv*) was supplied for this internship project. It contains **100,000 observations** and **14 fields** for **50 machines**, with dates running from 1 January to 10 March 2025 (69 calendar dates) and timestamps at one-minute resolution. Each record is a single reading for one machine; consecutive records are one minute apart but belong to different machines, so an individual machine is observed at irregular intervals (mean gap about 49 minutes; Section 4.4). Table 2 lists the fields. The supplied materials do not document how the measurements were collected or which network technology they describe; "6G" in the project title therefore frames the motivation rather than identifying a field in the data.

Table 2. Dataset Field Inventory

Field	Type	Meaning	Observed range / categories
Date	Text → date	Calendar date (dd-mm-yyyy)	1 Jan – 10 Mar 2025 (69 dates)
Timestamp	Text → time	Time of day of the observation	00:00:00 – 23:59:00 (1,440 values)
Machine_ID	Integer	Machine identifier	1 – 50 (50 machines)
Operation_Mode	Categorical	Operating mode	Active, Idle, Maintenance
Temperature_C	Numeric	Temperature (°C)	30.0 – 89.999
Vibration_Hz	Numeric	Vibration (Hz)	0.1 – 5.0
Power_Consumption_kW	Numeric	Power consumption (kW)	1.5 – 10.0
Network_Latency_ms	Numeric	Network latency (ms)	1.0 – 50.0
Packet_Loss_%	Numeric	Packet loss (%)	0.0 – 5.0
Quality_Control_Defect_Rate_%	Numeric	Quality-control defect rate (%)	0.0 – 10.0
Production_Speed_units_per_hr	Numeric	Production speed (units per hour)	50.0 – 499.997
Predictive_Maintenance_Score	Numeric	AI-derived maintenance readiness score	0.0 – 1.0
Error_Rate_%	Numeric	Operational error rate (%)	0.0 – 15.0
Efficiency_Status	Categorical	Efficiency category	Low, Medium, High

Ranges are the observed minimum and maximum in the raw dataset.

3.2 Data Quality Assessment

The raw file was audited before any transformation (no records were altered during the audit). Table 3 reports each check actually performed. The dataset has no missing values and no fully duplicated rows, every numeric measure lies within its plausible range, and the three operation modes and three efficiency classes are the only categories present. One structural anomaly was found and is described below.

Table 3. Data-Quality Audit Summary

Check	Result	Treatment
Dimensions	100,000 rows × 14 fields	—
Missing values	0 in every field	None required
Fully duplicated rows	0	None removed
Machine coverage	50 distinct Machine_ID values	—
Date continuity	69 dates; every gap between consecutive dates is 1 day	—
Records per date	67 dates with 1,440; 1 Mar 2025 with 2,880; 10 Mar 2025 with 640 (ends 10:39)	1 Mar flagged (below)
Repeated Date + Timestamp pairs	1,440 pairs, all on 1 Mar 2025	Retained; flagged
Repeated Date + Timestamp + Machine_ID	28 combinations	Retained; flagged
Chronological order	One backward step (1 Mar 2025, 00:00, start of second block)	Documented
Negative values	None in the nine numeric measures	—
Range validity	Packet loss, defect, and error rates within 0–100%; maintenance score within 0–1; 0 violations	—
Categories	3 operation modes; 3 efficiency classes; none unexpected	—

Timestamp anomaly (1 March 2025). The date 1 March 2025 contains 2,880 observations — twice the 1,440 found on other complete dates. Its timestamp sequence consists of two consecutive, identical 1,440-minute sequences, which produces the single backward chronological step. Both blocks contain all 50 machines, but the machine order differs between them (the machine sequences are not equal), and no row is a full duplicate of another; block means also differ (for example, mean latency 26.0 ms versus 25.2 ms, and mean temperature 60.05 °C versus 58.44 °C). Because the extra records are therefore distinct observations rather than copies, they could not be removed on the available evidence. They were **retained** and flagged with a Boolean indicator, and the anomaly is treated as a limitation on interpreting timestamps as a continuous real-time sequence.

3.3 Data Preparation

The raw dataset was loaded once and preserved unchanged; all work used a copy. Date strings were parsed (day-month-year) and timestamps converted to time-of-day values, from which a combined date-time field was built. The 1 March 2025 records were flagged, and **no rows were removed, imputed, or capped**. Two derived fields were then added — the Network Stability Index and its tertile-based Network Quality group (Sections 3.5–3.6). Finally, validation assertions confirmed 100,000 observations, 50 machines, no missing latency, packet-loss, or efficiency values, latency within 0–50 ms, packet loss within 0–5%, maintenance score within 0–1, only the three expected efficiency classes, an efficiency distribution that reconciles to the total, and finite KPI values. All analyses use the full cleaned dataset, including the 1 March 2025 records.

3.4 Network Performance Profiling

Latency and packet loss were summarized by count, mean, median, standard deviation, minimum, maximum, and the 5th, 10th, 25th, 50th, 75th, 90th, and 95th percentiles, and displayed as histograms with kernel density curves. Temporal patterns were examined descriptively at three levels: daily means and within-day spread (69 dates); clock-time means (1,440 distinct timestamps); and the daily mean of the Network Stability Index. Because each machine is observed at irregular intervals, these summaries are not treated as a continuous real-time network stream.

3.5 Network Quality Classification

Observations were divided into three equal-frequency groups (tertiles) of the Network Stability Index, labeled Low, Medium, and High network quality, where *Low* denotes the least stable third of observations. The tertile cut-offs are 0.409 and 0.591, producing groups of 33,334, 33,333, and 33,333 observations. These groups are **relative to this dataset**; they are not engineering categories and should not be read as 6G performance classes.

3.6 Network Stability Index

The Network Stability Index (NSI) combines the two available network indicators into a single 0–1 value. Latency and packet loss were included because they are the only two network-performance measurements in the data and both indicate degradation as they rise. Each was min–max normalized over the full dataset (latency 1–50 ms; packet loss 0–5%), the two normalized values were averaged with equal weights, and the result was inverted so that **higher values indicate more stable network conditions**:

$$NSI = 1 - (0.5 \times L_{norm} + 0.5 \times PL_{norm})$$

$$L_{norm} = (L - L_{min}) \div (L_{max} - L_{min}),\ PL_{norm} = (PL - PL_{min}) \div (PL_{max} - PL_{min})$$

Equal weighting was chosen because no weighting scheme was specified; it is an analytical assumption, not an established engineering standard. The index is unitless, ranges from 0.001 to 0.999 in the data, and has a mean of 0.5001. Because it is normalized to this dataset's own minimum and maximum, its values and the tertile thresholds derived from it are dataset-relative and are not comparable with other datasets.

3.7 Network–Efficiency Analysis

Efficiency_Status is categorical (Low, Medium, High) and was kept categorical: the share of each class was computed within each network-quality group (and within latency deciles) and compared directly. The classes were not coded 1–3 and averaged, because that would impose a numeric spacing between categories that the data do not justify. Production speed, a continuous measure, was used as a supporting indicator.

3.8 Latency Impact Diagnostics

Three related but different tools were used. (i) The **Spearman rank correlation** (Spearman, 1904) between latency and production speed measures the strength and direction of a monotonic (consistently rising or falling) relationship without assuming linearity. (ii) A **simple linear regression slope** (least-squares fit of production speed on latency) expresses the descriptive change in units per hour per millisecond. (iii) **Decile comparisons** divide observations into ten equal-frequency latency groups (D1 lowest to D10 highest, about 10,000 observations each) and compare mean and median production speed and the efficiency mix. A *sensitivity zone* was defined as a sustained, directional deterioration across successive latency ranges, as opposed to an isolated deviation in a single decile.

Because the data contain no explicit indicator of real-time versus delayed communication, a descriptive **observation-gap proxy** was also examined: for each machine, the time since its previous observation was computed, and observations with gaps at or below the 25th percentile (14 minutes) were compared with those at or above the 75th percentile (68 minutes), excluding the middle half. This compares shorter and longer observation intervals; it does not measure communication delay.

3.9 Packet Loss Impact Diagnostics

The Spearman correlation between packet loss and error rate was computed, and packet-loss deciles were used to compare mean defect rate (and median), mean error rate, and mean production speed across increasing packet-loss ranges. The **Packet Loss Impact Ratio** contrasts mean production speed in the lowest and highest packet-loss deciles. It describes an observed difference between two extreme groups; it is not an estimate of damage caused by packet loss. It is computed from the *unrounded* decile means (278.0986 and 273.8100 units/hr), which gives 1.5421%; the rounded values shown in the tables (278.10 and 273.81) would give 1.5426%, so the reported figure should always be taken from the unrounded means. A packet-loss *risk zone* is defined analogously to a latency sensitivity zone: a sustained, directional deterioration across successive packet-loss ranges.

3.10 Operation Mode Interaction

Network and outcome indicators were compared across Active, Idle, and Maintenance modes (observation counts, mean latency, packet loss, production speed, error rate, and defect rate). To check whether any mode responds differently to latency, mean production speed was computed within each mode across the dataset-wide latency deciles, and the three profiles were compared for any sustained directional change.

3.11 Comparable Mechanical Conditions

Temperature, vibration, and power consumption were each divided into quartile bands (Low, Medium-Low, Medium-High, High). Their combination defines $4 \times 4 \times 4 = 64$ mechanical-condition profiles, each containing observations with similar mechanical characteristics. Within every profile, the efficiency mix was computed separately for each network-quality group, and the results were summarized across profiles (mean, standard deviation, minimum, maximum). The purpose is to check whether a network-related pattern becomes visible once temperature, vibration, and power are held within comparable ranges. This is an **approximation** to a comparison under identical mechanical conditions, not exact matching: observations in a profile share broad ranges of temperature, vibration, and power rather than equal values. It **reduces but does not eliminate** confounding; other factors, such as workload or machine identity, remain uncontrolled, and the bands are dataset-relative rather than machine-health limits. For this reason the paper refers throughout to comparison *within comparable mechanical-condition profiles*.

3.12 KPI Definitions

Four KPIs summarize the analysis (Table 4). Each is computed once in the analysis notebook and reused unchanged here and in the dashboard.

Table 4. Key Performance Indicators: Definitions and Interpretation Boundaries

KPI	Definition and calculation	Unit	Interpretation boundary
Network Stability Index	1 − (0.5 × normalized latency + 0.5 × normalized packet loss); min–max normalization over the dataset	Unitless, 0–1 (higher = more stable)	Dataset-relative; equal weighting is an assumption
Latency Sensitivity Score	Slope of the least-squares line of production speed on latency	Units/hr per ms	Descriptive slope, not a causal effect
Packet Loss Impact Ratio	(Mean speed in packet-loss decile D1 − mean speed in D10) ÷ mean speed in D1 × 100	Percent	Extreme-decile contrast, not causal degradation
Network–Efficiency Relationship (assignment KPI: "Network-Efficiency Correlation")	Assessment of whether any network range is linked to a sustained efficiency decline across network-quality groups, latency deciles, packet-loss ranges, and comparable mechanical-condition profiles	Qualitative (no coefficient)	Reported as "Not identified"; no drop point or threshold is inferred

Mapping to the assignment KPI. The assignment names this indicator *Network-Efficiency Correlation* and describes it as the network point at which efficiency drops. Efficiency_Status is categorical (Low, Medium, High) and is deliberately not coded as numbers (Section 3.7), so a single correlation coefficient would not be meaningful. The indicator is therefore implemented as the evidence-based assessment in the last row of Table 4: efficiency mix was compared across network-quality groups, latency deciles, packet-loss ranges, and comparable mechanical-condition profiles, and no point at which efficiency drops was identified. Accordingly, no coefficient or threshold is reported for this KPI, and none was fabricated.

3.13 Analytical Tools and Reproducibility

The analysis was implemented in Python within a Jupyter notebook using pandas (McKinney, 2010), NumPy (Harris et al., 2020), Matplotlib (Hunter, 2007), and seaborn (Waskom, 2021). The notebook preserves the raw data, documents each cleaning decision, ends with validation assertions, and exports the saved figure and table files traced in Appendix A. The Streamlit dashboard reproduces these analyses interactively (tabs: Overview, Network × Efficiency, Latency Impact, Quality & Errors, Operation Modes, 6G Insights, and Method & Limits). For speed, the dashboard draws row-level scatter plots from a fixed-seed random sample of 6,000 points while computing every statistic from all selected observations. In this paper, the scatter panels show every observation, and all statistics use the full dataset.

4. RESULTS

This section reports what the analysis found; interpretation is reserved for Section 5. All results are descriptive associations in this dataset.

4.1 Dataset Overview and Quality

The cleaned dataset contains 100,000 observations from 50 machines over 69 dates (1 January to 10 March 2025), with no missing values and no full-row duplicates (Section 3.2). The 1 March 2025 anomaly (2,880 records on that date) was retained and flagged. Table 5 gives the efficiency and operation-mode composition. The efficiency classes are **highly imbalanced**: Low accounts for 77.83% of observations, Medium for 19.19%, and High for only 2.99% (2,986 observations). This matters for every later comparison: with a baseline of nearly 78% Low, a network-related shift would have to be judged against that baseline, and percentages for the High class rest on comparatively few observations.

Table 5. Composition of the Dataset by Efficiency Class and Operation Mode

Efficiency class	Observations	Share	Operation mode	Observations	Share
Low	77,825	77.83%	Active	70,054	70.05%
Medium	19,189	19.19%	Idle	20,057	20.06%
High	2,986	2.99%	Maintenance	9,889	9.89%
Total	100,000	100.00%	Total	100,000	100.00%

Shares are rounded to two decimals; the rounded efficiency shares (77.83%, 19.19%, 2.99%) total 100.01% although the underlying counts total exactly 100,000.

4.2 Network Performance Profile (RQ1)

Latency averaged 25.56 ms (median 25.54 ms, standard deviation 14.12 ms) and ranged from 1.00 to 50.00 ms; packet loss averaged 2.49% (median 2.49%, standard deviation 1.44%) and ranged from 0.00% to 5.00% (Table 6). The distributions are close to flat across these bounded ranges (Figure 2). Their percentiles closely match those expected of a uniform spread: for example, the 5th and 95th latency percentiles are 3.50 and 47.58 ms (3.45 and 47.55 ms would be expected for a uniform distribution on 1–50 ms), and the corresponding packet-loss percentiles are 0.249% and 4.755% (0.25% and 4.75% expected for 0–5%). Two consequences follow: the data contain no heavy tail of extreme network degradation, and the observed range (up to 50 ms and 5% loss) is the full range available for analysis.

Table 6. Summary Statistics of Network Indicators

Measure	Mean	Median	Std. dev.	Min	Q1	Q3	Max
Latency (ms)	25.556	25.536	14.121	1.000	13.355	37.796	50.000
Packet loss (%)	2.493	2.488	1.443	0.000	1.245	3.741	5.000
Network Stability Index	0.500	0.500	0.203	0.001	0.355	0.645	0.999

Q1 and Q3 are the 25th and 75th percentiles.

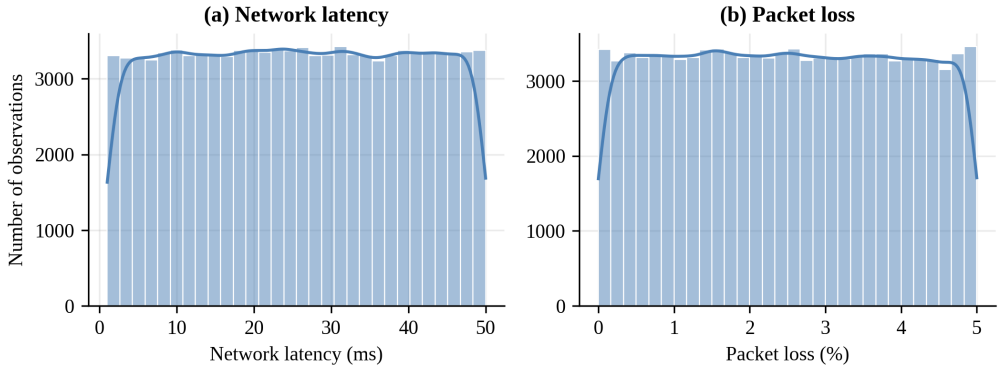


Figure 2. Distribution of (a) network latency and (b) packet loss across all 100,000 observations, with kernel density curves.

Temporal variation is small. Daily mean latency ranged from 24.69 to 26.50 ms (standard deviation of daily means 0.34 ms) and daily mean packet loss from 2.42% to 2.60%; the daily mean NSI stayed between 0.481 and

0.512 (Figure 3). Across the 1,440 clock minutes, mean latency ranged from 20.26 to 31.61 ms, which is the scale expected from sampling alone: with roughly 69 observations per clock minute, independent draws from a stable distribution with a standard deviation of 14.12 ms would give a standard deviation of means of about 1.7 ms, and the observed value is 1.71 ms. No distinct date-level or time-of-day period of sustained network stability or instability is therefore identified.

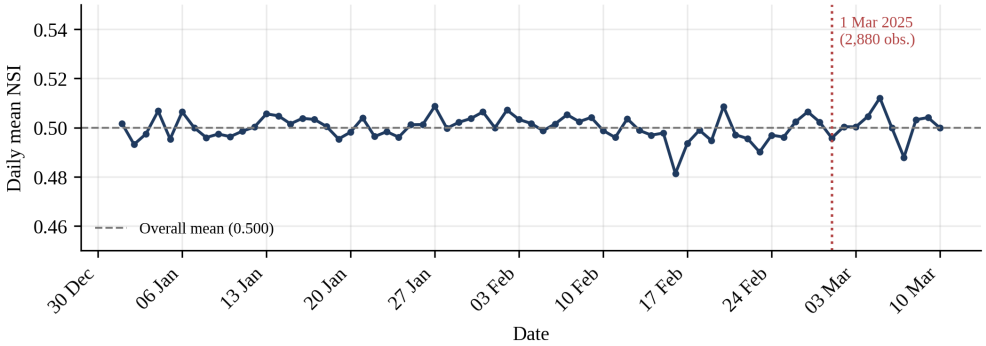


Figure 3. Daily mean Network Stability Index. The vertical axis is deliberately limited to 0.45–0.55 to make small day-to-day variation visible; the index itself spans 0–1. The dotted line marks 1 March 2025, the date with 2,880 records.

Table 7 describes the three network-quality groups. They are clearly separated on the underlying measures — mean latency falls from 36.63 ms (Low) to 25.63 ms (Medium) and 14.41 ms (High), and mean packet loss from 3.63% to 2.49% and 1.36% — so the grouping genuinely contrasts poorer and better network conditions within the data. Each group contributes about one-third of the observations on every date (date-level standard deviation about 0.01) and all three groups occur at every clock time, so the groups are not an artifact of a few dates or hours.

4.3 Network Quality and Efficiency (RQ2)

The efficiency mix is essentially the same in all three network-quality groups (Table 7, Figure 4). Low efficiency accounts for 77.89%, 77.75%, and 77.84% of the Low-, Medium-, and High-quality groups, respectively — a maximum difference of 0.14 percentage points; Medium efficiency ranges from 19.08% to 19.33%, and High efficiency from 2.92% to 3.03%. Across the ten latency deciles, Low efficiency varies between 77.35% and 78.49% (a 1.14-point spread), Medium between 18.54% and 19.52%, and High between 2.65% and 3.29%, with no sustained rise or fall in any class as latency increases (Table 8). The better-quality groups do not show a higher share of High efficiency, and the poorer-quality group does not show a higher share of Low efficiency.

Table 7. Network-Quality Groups: Composition, Network Conditions, and Efficiency Mix

Group	Obs.	NSI range	Mean latency (ms)	Mean packet loss (%)	Mean NSI	Low eff. (%)	Medium eff. (%)	High eff. (%)
Low quality	33,334	0.001–0.409	36.63	3.63	0.273	77.89	19.08	3.03
Medium quality	33,333	0.409–0.591	25.63	2.49	0.500	77.75	19.33	2.92
High quality	33,333	0.591–0.999	14.41	1.36	0.727	77.84	19.16	3.01

"Low quality" means the least stable third of observations. Efficiency columns show the share of each efficiency class within the group.

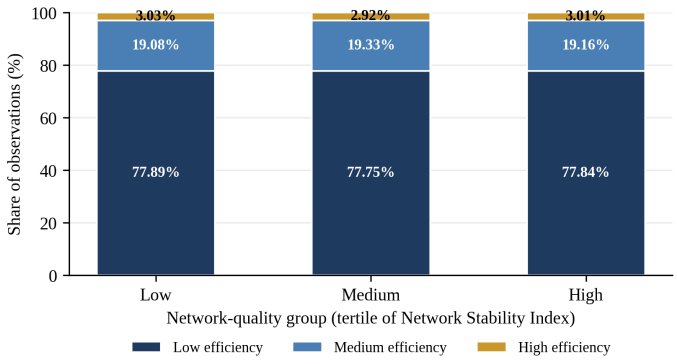


Figure 4. Distribution of efficiency status within each dataset-relative network-quality group (percentage of observations).

4.4 Latency and Production Speed (RQ3)

The Spearman correlation between latency and production speed is **−0.0011**, an essentially null monotonic association. The fitted linear slope (the Latency Sensitivity Score) is **−0.00967** units/hr per millisecond. To put that in perspective, across the entire observed latency range of 49 ms the fitted line falls by only about 0.47 units/hr, roughly 0.17% of the mean production speed of 275.92 units/hr. Figure 5(a) shows production speeds spread across their full range at every latency, with a nearly flat fitted line.

The decile analysis (Table 8, Figure 5(b)) agrees. Mean production speed rises from 275.11 units/hr in D1 to 278.52 in D5, drops to its lowest value, 272.81 units/hr, in D6 (latency 25.54–30.42 ms), recovers to 277.97 in D8, and is 274.56 in the highest-latency decile, D10 (45.13–50.00 ms), which is not the lowest. The D6 dip is therefore isolated, not part of a sustained decline. The total spread of decile means (272.81 to 278.52, about 5.7 units/hr or 2.1% of the mean) is small relative to the overall standard deviation of production speed (130.10 units/hr). **No distinct, sustained latency sensitivity zone is identified.**

Table 8. Production Speed and Efficiency Mix Across Latency Deciles

Decile	Latency range (ms)	Mean speed (units/hr)	Median speed (units/hr)	Low eff. (%)	Medium eff. (%)	High eff. (%)
D1	1.00–5.98	275.11	275.63	77.93	19.13	2.94
D2	5.98–10.89	275.46	275.34	77.98	19.00	3.02
D3	10.89–15.83	276.24	276.67	77.73	19.37	2.90
D4	15.83–20.70	278.03	279.55	77.35	19.51	3.14
D5	20.70–25.54	278.52	281.84	77.48	19.23	3.29
D6	25.54–30.42	272.81	271.88	78.49	18.54	2.97
D7	30.42–35.30	274.36	274.86	77.57	19.52	2.91
D8	35.30–40.24	277.97	279.22	77.65	19.24	3.11
D9	40.24–45.12	276.09	276.26	78.13	19.22	2.65
D10	45.13–50.00	274.56	275.31	77.94	19.13	2.93

Deciles are equal-frequency groups of about 10,000 observations; D1 = lowest latency.

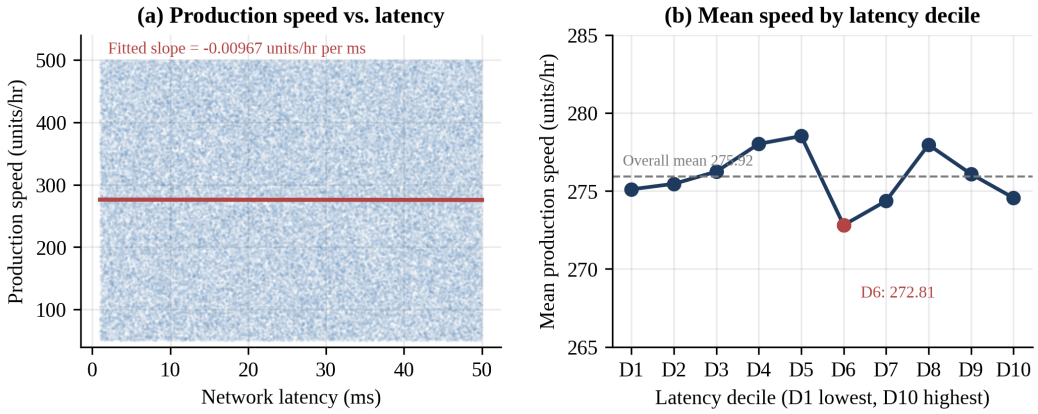


Figure 5. Production speed and latency: (a) all 100,000 observations with the fitted least-squares line; (b) mean production speed by latency decile (vertical axis limited to 265–285 units/hr).

Observation-gap proxy. Per-machine gaps between consecutive observations averaged 49.25 minutes (median 34, quartiles 14 and 68, maximum 514); 28 gaps (0.03%) were zero, corresponding to the repeated machine–timestamp combinations on 1 March 2025. Comparing the short-gap group (≤ 14 minutes; 25,065 observations) with the long-gap group (≥ 68 minutes; 25,269 observations) shows near-identical mean production speed (275.39 versus 275.14 units/hr), latency (25.56 versus 25.48 ms), and packet loss (2.50% versus 2.49%), and a similar efficiency mix (Low 78.02% versus 77.57%). Because gaps are not measurements of communication delay, this establishes nothing about real-time versus delayed communication; it simply shows that shorter and longer sampling intervals coincide with similar outcomes.

4.5 Packet Loss and Error Rate (RQ4)

The Spearman correlation between packet loss and error rate is **−0.0024**, an essentially null monotonic association, far below the value of about 0.1 conventionally used to label even a "small" association (Cohen, 1988). The fitted line in Figure 6(a) is nearly flat. Mean error rate across packet-loss deciles stays within 7.47%–7.56% (Table 9). The negative sign should not be read as packet loss *improving* reliability: a coefficient this close to zero gives no evidence of any meaningful direction.

4.6 Packet Loss and Defect Rate (RQ5)

Mean defect rate remains between 4.95% and 5.07% across all ten packet-loss deciles — a total spread of 0.12 percentage points — with no sustained increase toward higher packet loss (Table 9, Figure 6(b)). The highest decile (packet loss 4.50–5.00%) has a mean defect rate of 5.01%, indistinguishable from the lowest-packet-loss deciles (5.02% in D1). Production speed varies more visibly: it is highest in D1 (278.10 units/hr) and lowest in D10 (273.81), but the intermediate deciles move up and down (for example, 274.26 in D6 but 276.94 in D5 and 276.73 in D9), so there is no consistent progressive decline. No packet-loss range is identified as a defect, error, speed, or efficiency threshold.

Table 9. Production Speed, Error Rate, and Defect Rate Across Packet-Loss Deciles

Decile	Packet-loss range (%)	Mean speed (units/hr)	Mean error rate (%)	Mean defect rate (%)	Median defect rate (%)
D1	0.00–0.50	278.10	7.51	5.02	5.00
D2	0.50–1.00	277.64	7.50	5.07	5.12
D3	1.00–1.50	276.31	7.51	5.04	5.06
D4	1.50–1.99	275.07	7.50	5.01	5.01
D5	1.99–2.49	276.94	7.53	5.00	5.02
D6	2.49–2.99	274.26	7.56	4.98	4.94
D7	2.99–3.49	275.23	7.48	4.95	4.95
D8	3.49–3.99	275.07	7.49	4.98	4.94
D9	3.99–4.50	276.73	7.47	5.03	5.02
D10	4.50–5.00	273.81	7.50	5.01	5.00

Deciles are equal-frequency groups of about 10,000 observations; D1 = lowest packet loss.

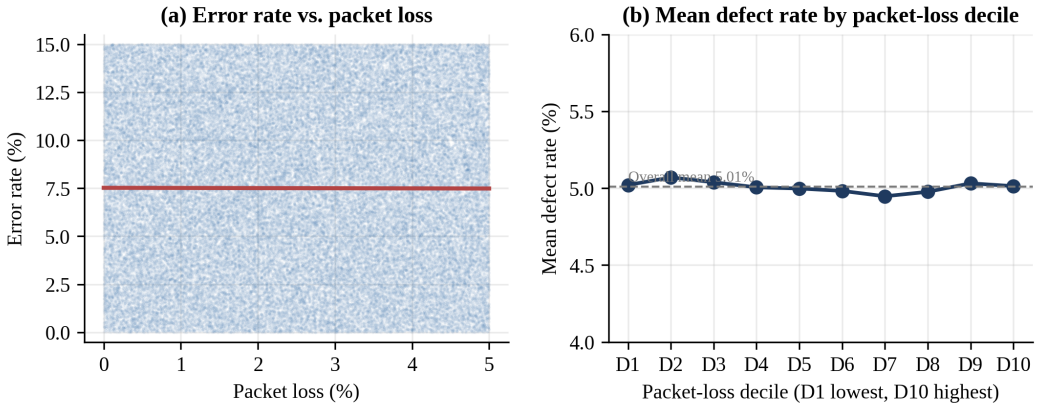


Figure 6. Packet loss and quality outcomes: (a) error rate versus packet loss for all 100,000 observations with the fitted line; (b) mean defect rate by packet-loss decile (vertical axis limited to 4–6%).

4.7 Operation Mode Interaction (RQ6)

Network conditions are almost identical across the three modes (Table 10): mean latency ranges from 25.50 to 25.64 ms and mean packet loss from 2.48% to 2.50%. Outcomes differ little: mean production speed is 276.20 units/hr (Active), 274.13 (Idle), and 277.54 (Maintenance), a total spread of 3.41 units/hr (about 1.2%), while error rates (7.49%–7.54%) and defect rates (5.00%–5.01%) are practically equal. The modest speed differences between modes should therefore not be attributed to network conditions.

Table 10. Network and Manufacturing Indicators by Operation Mode

Operation mode	Observations	Mean latency (ms)	Mean packet loss (%)	Mean speed (units/hr)	Mean error rate (%)	Mean defect rate (%)
Active	70,054	25.54	2.49	276.20	7.49	5.01
Idle	20,057	25.64	2.50	274.13	7.52	5.01
Maintenance	9,889	25.50	2.48	277.54	7.54	5.00

Within each mode, production speed across the latency deciles fluctuates without a sustained decline (Figure 7). Active operation peaks at D5 (280.71 units/hr), dips at D6 (272.88), and recovers; Idle operation is lowest in D10 (268.86) but also reaches its highest value in D2 (280.39); Maintenance alternates up and down (271.34 in D7, 282.42 in D8). The Idle and Maintenance profiles are noisier, but they are built from far fewer observations (about 2,000 and 1,000 per decile, against about 7,000 for Active), and smaller groups naturally produce more volatile means. No operation mode is identified as having a distinct latency-sensitive production zone.

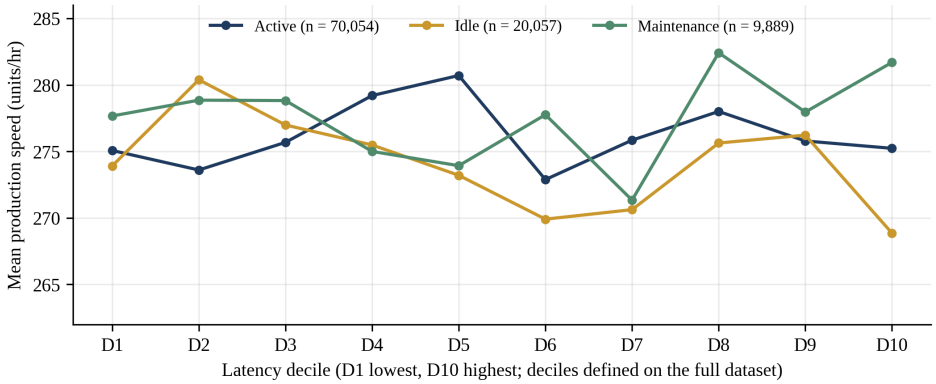


Figure 7. Mean production speed across latency deciles by operation mode (vertical axis limited to 262–286 units/hr).

4.8 Comparable Mechanical Conditions (RQ7)

The 64 mechanical-condition profiles are well populated: profile sizes range from 1,492 to 1,640 observations (mean 1,562.5), so no profile is sparse. Within a profile, each of the three network-quality groups contains about 521 observations (range 465–595). When efficiency mix is compared across network-quality groups *within* these comparable profiles (an approximation to equal mechanical conditions, not exact matching) and averaged (Table 11, Figure 8), the pattern is unchanged: the mean share of Low efficiency is 77.88%, 77.76%, and 77.84% for Low-, Medium-, and High-quality networks; Medium efficiency 19.08%, 19.32%, and 19.15%; and High efficiency 3.04%, 2.92%, and 3.00%. Differences between network-quality groups (at most 0.24 percentage points) are small compared with the profile-to-profile variation (standard deviation 1.5–1.9 points for the Low and Medium classes). No operation mode shows a consistently distinct efficiency pattern across network-quality categories in this comparison. Two cautions apply: each profile-by-group cell is small (so individual percentages, especially for the High class, which has only about 15 observations per cell, are noisy), and grouping by comparable temperature, vibration, and power bands does not control for every other factor.

Table 11. Efficiency Mix by Network Quality Within Comparable Mechanical-Condition Profiles (64 Profiles)

Network quality	Efficiency class	Mean (%)	Std. dev. (points)	Min (%)	Max (%)
Low	Low	77.88	1.64	73.95	81.54
Low	Medium	19.08	1.50	15.58	23.76
Low	High	3.04	0.77	1.62	4.97
Medium	Low	77.76	1.79	72.92	81.95
Medium	Medium	19.32	1.66	14.40	23.79
Medium	High	2.92	0.72	1.56	4.61
High	Low	77.84	1.94	72.64	82.88
High	Medium	19.15	1.77	14.80	23.46
High	High	3.00	0.80	1.63	5.39

Each value summarizes, across the 64 profiles, the percentage of observations in the given efficiency class within a profile and network-quality group.

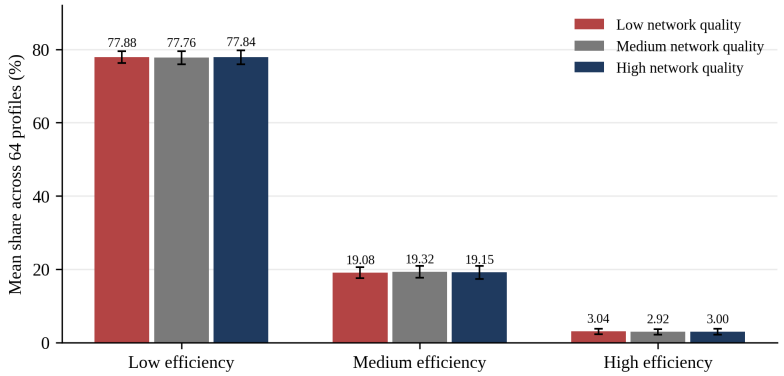


Figure 8. Mean efficiency mix by network-quality group across 64 comparable mechanical-condition profiles (error bars: standard deviation across profiles). Derived from the notebook's comparable-condition summary table.

4.9 KPI Results

Table 12 consolidates the final source-of-truth metrics. The Network Stability Index averages 0.5001; the Latency Sensitivity Score is −0.00967 units/hr per ms; and the Packet Loss Impact Ratio is 1.5421%, i.e., mean production speed was 278.0986 units/hr in the lowest packet-loss decile and 273.8100 in the highest (278.10 and 273.81 when rounded). Because D1 and D10 are, respectively, the highest and lowest of all ten decile means, the ratio is the widest gap among the packet-loss decile means (and is not a measure of a progressive decline). The Network–Efficiency Relationship (the assignment's Network-Efficiency Correlation, Section 3.12) is reported as **Not identified** for latency, packet loss, and network-quality shifts.

Table 12. Final Source-of-Truth Metrics

Metric	Final value
Total observations	100,000
Number of machines	50
Date range	2025-01-01 to 2025-03-10
Mean / median latency (ms)	25.5556 / 25.5360
Mean / median packet loss (%)	2.4934 / 2.4880
Low efficiency	77,825 (77.83%)
Medium efficiency	19,189 (19.19%)
High efficiency	2,986 (2.99%)
Mean Network Stability Index	0.5001
Latency Sensitivity Score	−0.00967 units/hr per ms
Packet Loss Impact Ratio	1.5421%
Network-efficiency sensitivity finding	No distinct sensitivity zone identified

4.10 Latency Tolerance and Packet-Loss Risk-Zone Assessment

The assignment requests latency tolerance benchmarks and packet-loss risk zones. This subsection states what was examined, what was observed, and what can and cannot be concluded (Table 13).

Latency tolerance. The full observed latency range (1–50 ms) was examined in ten equal-frequency deciles, in three network-quality groups, and within each operation mode, against production speed and efficiency mix (Sections 4.3, 4.4, 4.7). No range showed a sustained deterioration: the only notable dip (D6, 25.54–30.42 ms, 272.81 units/hr) was followed by recovery, and the highest-latency decile (45.13–50.00 ms) was not the lowest-performing one. **Packet-loss risk zones.** The observed range (0–5%) was examined in ten deciles against production speed, error rate, and defect rate (Sections 4.5–4.6). Speed ranged from 273.81 to 278.10 units/hr without a consistent progression, mean error rate from 7.47% to 7.56%, and mean defect rate from 4.95% to 5.07%; the highest decile (4.50–5.00%) was no worse than the others on errors or defects.

Assessment. Within the observed ranges, no latency level or packet-loss band was associated with a consistent decline in production, rise in errors, or rise in defects. **No validated latency-tolerance benchmark or packet-loss risk zone could therefore be established from this dataset.** The mean values (25.56 ms and 2.49%)

describe the data and are not tolerance limits. Absence of an observed deterioration point is not proof that none exists: the observed ranges are bounded (nothing is known about latency above 50 ms or loss above 5%), the data are observational and aggregated, outcomes cannot be tied to specific communication links, the efficiency classes are imbalanced, and the dataset's provenance is undocumented (Sections 5.4 and 7). Any threshold drawn from these data would therefore be arbitrary. The defensible next steps are trend-based monitoring (Section 6) and validation of candidate thresholds on richer, synchronized operational data and relevant engineering standards (Section 7).

Table 13. Latency-Tolerance and Packet-Loss Risk-Zone Assessment

Indicator	Ranges examined	Outcomes compared	Observed pattern	Assessment
Latency	1–50 ms; deciles D1–D10; 3 network-quality groups; 3 operation modes	Production speed; efficiency mix	Decile means 272.81–278.52 units/hr with an isolated D6 dip; Low-efficiency share 77.35%–78.49%; no sustained decline	No sensitivity zone; no tolerance benchmark established
Packet loss	0–5%; deciles D1–D10	Production speed; error rate; defect rate	Speed 273.81–278.10 units/hr (non-monotonic); errors 7.47%–7.56%; defects 4.95%–5.07%	No risk zone identified

Results describe the observed ranges only and are not engineering thresholds.

Table 14. Summary of Findings by Research Question

RQ	Key evidence	Finding
RQ1	Latency 1–50 ms (mean 25.56); packet loss 0–5% (mean 2.49); daily NSI 0.481–0.512	Bounded, near-uniform spread; no distinct stable or unstable periods
RQ2	Low-efficiency share 77.75%–77.89% across network-quality groups	Efficiency mix essentially unchanged across groups
RQ3	Spearman −0.0011; slope −0.00967 units/hr per ms; decile means 272.81–278.52	No meaningful association; no sustained sensitivity zone
RQ4	Spearman −0.0024; error rate 7.47%–7.56% across deciles	Essentially no association
RQ5	Defect rate 4.95%–5.07% across deciles; 5.01% in the highest decile	Defect rate stable across packet-loss ranges
RQ6	Network conditions near-identical across modes; no sustained latency decline in any mode	No mode identified as distinctly sensitive
RQ7	Low-efficiency share 77.76%–77.88% across groups within 64 profiles	Pattern unchanged under comparable conditions
RQ8	No threshold, tolerance benchmark, or risk zone identified in any analysis (Section 4.10)	Supports trend-based monitoring; thresholds need further validation

5. DISCUSSION

5.1 Interpretation of the Findings

Across every comparison, the observed relationships between network performance and manufacturing outcomes are weak or absent in this dataset. Correlations are near zero (−0.0011 and −0.0024), efficiency mix is stable across network groups and latency deciles, defect and error rates are flat across packet-loss ranges, and the same holds within operation modes and comparable mechanical profiles. Efficiency differences of at most 0.14 percentage points between network groups are also smaller than the sampling variation expected for a proportion near 78% estimated from about 33,000 observations (standard error of roughly 0.2 points; an approximate benchmark for context, not a formal test), so they should not be read as signals. The few visible features — the D6 latency dip, the 1.54% extreme-decile contrast — are isolated and non-monotonic, and the notebook's conclusion that they do not constitute a sensitivity zone is supported by the decile profiles.

A weak relationship is a valid finding, but it is a finding about *this dataset*. It does not show that latency or packet loss never affect manufacturing; it shows that, within the observed ranges and measurement design, no consistent association was found.

5.2 Operational Implications

The practical implication is that network indicators should not be used on their own to explain changes in production, errors, or quality: a deterioration in output observed in this kind of data would not be well explained by latency or packet loss alone. Network measurements remain worth collecting as one input to diagnosis, interpreted alongside operation mode and machine condition, because their absence from a diagnosis cannot be justified by this dataset and their relevance in other conditions is supported by the general literature. Because no stable sensitivity threshold emerged, changes and persistent deviations are a more defensible trigger for investigation than a fixed cut-off derived from these data.

5.3 Comparison with Existing Literature

The networked-control literature shows that delay and packet loss can degrade the stability and performance of feedback control under modeled conditions (Zhang et al., 2001; Hespanha et al., 2007; Schenato et al., 2007), and the 3GPP requirements emphasize reliability and low latency for cyber-physical control (3GPP, TS 22.104). The present findings are not in conflict with this work. Those results concern specific control loops, explicit models of loss and delay, and defined protocols; the present dataset records aggregate outcome measures (production speed, error rate, defect rate, efficiency class) without identifying which communication links, if any, those outcomes depend on. A near-zero association in such aggregate data is consistent with the literature if, for example, the monitored communication does not sit in a performance-critical loop or if the impairments observed (up to 50 ms and 5% loss) lie within the tolerance of the processes concerned. The observed ranges should not be equated with the ultra-reliable, very-low-latency service levels discussed for demanding control applications, and the dataset does not state whether its measurements pertain to such traffic.

5.4 Why Expected Relationships May Not Appear

Several explanations are possible; none is established by this analysis. (i) Network performance may not be the dominant source of variation: production speed spans 50 to 500 units/hr within every network group, so other factors dominate the variance. (ii) Mechanical and operational conditions may drive outcomes, although grouping by comparable temperature, vibration, and power did not reveal a hidden network effect. (iii) The observed ranges may not include severe degradation. (iv) The measurements may not represent the communication relevant to control or quality. (v) Aggregated observational data can obscure local effects, such as those specific to one machine or event.

A further possibility concerns the data themselves. Several fields are spread almost uniformly over round-number ranges, and their summary statistics coincide with theoretical values: defect rate (mean 5.01, standard deviation 2.88), error rate (7.50, 4.34), and maintenance score (0.50, 0.29) match the values expected of uniform distributions over 0–10, 0–15, and 0–1 (midpoints 5, 7.5, 0.5; standard deviation = range ÷ √12 = 2.89, 4.33, 0.29). In addition, production speed in Idle and Maintenance modes is about the same as in Active mode. These regularities are unusual for field measurements and are typical of simulated or randomly generated data. The provenance of the dataset is not documented, so this is a caution rather than a conclusion: if the values were generated independently, the absence of relationships may reflect the generation process rather than the behavior of a real factory, and the data owner should be asked how the measurements were produced.

5.5 Implications for Network Monitoring

For monitoring, the findings suggest three principles: pair each network indicator with the outcome and condition variables needed to interpret it; watch for trends, persistent deviations, and unusual observations instead of relying on a universal threshold; and treat the Network Stability Index and its tertile groups as *relative* tools for comparing conditions within one system, not as specifications. Any production-grade threshold would require validation against richer operational data and against the engineering standards relevant to the processes concerned.

6. RECOMMENDATIONS

The recommendations below are derived from the analysis; they have not been implemented or validated in a live factory, and they rest on the supplied dataset only: this study does not independently validate a real 6G deployment, and the dataset's provenance is undocumented. For each, the supporting evidence, a possible action, and the main condition are given.

1. **Integrated monitoring.** Evidence: network indicators alone did not explain efficiency, speed, errors, or defects (Sections 4.3–4.6). Action: track latency and packet loss together with production speed, error rate, defect rate, efficiency class, and machine conditions in one view. Condition: signals should be time-synchronized (see recommendation 6).
2. **Trend-based alerts rather than fixed thresholds.** Evidence: no sustained latency or packet-loss sensitivity zone was found (Sections 4.4, 4.6). Action: alert on persistent deviations or shifts relative to a machine's own baseline, rather than adopting a cut-off from this dataset. Condition: baselines must be re-estimated on future data.
3. **Context-aware diagnostics.** Evidence: results were consistent across operation modes and comparable mechanical profiles (Sections 4.7–4.8), but these contexts matter for interpretation. Action: when investigating a change, read network measurements within the machine's operation mode and mechanical condition. Condition: grouping reduces, but does not remove, confounding.
4. **Communication-reliability investigation.** Evidence: aggregate packet loss showed no relationship with errors or defects, but the data cannot identify which links matter. Action: collect richer diagnostics (for example, per-link delay variation, retransmissions, and communication-failure events) to test whether specific conditions coincide with operational issues. Condition: requires instrumentation beyond the present fields.
5. **Efficiency-class tracking with attention to imbalance.** Evidence: Low efficiency accounts for 77.83% of observations and High for 2.99% (Section 4.1). Action: monitor the full efficiency distribution and judge changes against this baseline rather than against equal-class expectations. Condition: confirm with process owners what defines each class.
6. **Better data collection.** Evidence: no communication-timing label, irregular per-machine sampling, a repeated timestamp block, and undocumented provenance limit inference (Sections 3.2, 5.4). Action: record network, machine, production, and defect measurements at compatible resolution with consistent timestamps, and document how measurements are generated. Condition: feasibility depends on plant systems.
7. **Validation before deployment.** Evidence: all dataset-relative groups and indices are specific to this data. Action: validate any proposed rule against future operational data and appropriate engineering standards before use in production. Condition: thresholds from this study should not be used as engineering limits.

7. LIMITATIONS AND FUTURE WORK

7.1 Limitations

- **Observational data.** The analysis shows associations, not causation; unobserved factors may influence both network and outcome variables.
- **No real-time/delayed label.** The observation-gap proxy compares sampling intervals and is not a measure of communication delay.
- **Dataset-relative groups.** Network-quality tertiles, deciles, and mechanical bands are relative to this dataset; they are not engineering classes, and no universal latency or packet-loss threshold should be inferred.
- **Index assumptions.** The Network Stability Index uses equal weights and min–max normalization, both analytical choices.
- **Extreme-decile ratio.** The Packet Loss Impact Ratio compares the highest and lowest of the ten decile means and is not a causal degradation percentage.
- **Efficiency imbalance.** Low efficiency makes up 77.83% of observations and High only 2.99%, limiting power to detect shifts in the rarer classes.
- **Timestamp structure.** The repeated 1,440-minute block on 1 March 2025 (2,880 records) was retained and flagged; it limits reading timestamps as a continuous sequence.
- **Residual confounding.** Comparable mechanical profiles do not control for workload, machine identity, or other factors; profile-by-group cells are small (about 521 observations).
- **Bounded ranges.** Latency (1–50 ms) and packet loss (0–5%) cover limited ranges, so effects under more severe degradation cannot be assessed.

- **Provenance and technology.** The data do not identify the network technology or how they were generated, and several statistical regularities (Section 5.4) suggest possible simulation.
- **Tool and sampling note.** The dashboard draws scatter plots from a fixed-seed sample of 6,000 points for speed; every statistic uses all selected observations.

7.2 Future Work

The following are proposed extensions, not completed work: collecting data from more machines, production lines, and periods; adding direct reliability and communication-timing indicators; recording synchronized machine, network, and quality events; validating the observed patterns on future data; using controlled experiments or stronger causal designs where feasible; investigating behavior under more severe network degradation; and evaluating domain-standard thresholds before operational adoption.

8. CONCLUSION

This project examined the relationship between network performance and manufacturing outcomes in a smart-factory dataset of 100,000 observations from 50 machines, using descriptive and diagnostic analytics. It characterized latency and packet loss, compared efficiency patterns across dataset-relative network-quality groups, investigated latency–production and packet-loss–quality relationships, and examined operation modes and comparable mechanical conditions.

The monotonic relationships were extremely weak: −0.0011 between latency and production speed, and −0.0024 between packet loss and error rate. Decile analyses identified no sustained latency sensitivity zone; defect rates stayed between 4.95% and 5.07% across packet-loss ranges; efficiency mix was practically unchanged across network-quality groups (Low efficiency 77.75%–77.89%), and the same held across operation modes and within comparable mechanical profiles.

These results support monitoring network performance alongside operational and mechanical conditions, and avoiding unsupported causal attribution. The study evaluates the supplied smart-factory dataset only: it does not independently validate a real 6G deployment, its provenance is undocumented, and several statistical regularities suggest possible simulation. The analysis is observational, the dataset has documented structural limits, and the groups and indices are dataset-relative. No latency-tolerance benchmark or packet-loss risk zone could be established. Further synchronized data collection and validation are needed before production-grade thresholds are set or network performance is claimed to determine manufacturing efficiency.

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APPENDIX A: TRACEABILITY TO THE ANALYSIS NOTEBOOK

Every figure and table in this paper is traceable to an output of the frozen analysis notebook. Figures were redrawn from the same data and logic at 300 dpi with consistent formatting; scatter panels show all 100,000 observations, and the restricted vertical axes noted in the captions affect display only.

Paper element	Notebook output / saved file
Figure 2 (distributions)	F01_Network_Performance_Distributions.png
Figure 3 (daily NSI)	F02_Daily_Network_Stability_Index.png
Figure 4 (efficiency by network quality)	F03_Efficiency_by_Network_Quality.png
Figure 5 (latency)	F04_Production_Speed_vs_Latency.png; F05_Production_Speed_by_Latency_Decile.png
Figure 6 (packet loss)	F06_Error_Rate_vs_Packet_Loss.png; F07_Defect_Rate_by_Packet_Loss_Range.png
Figure 7 (operation modes)	F08_Production_Speed_by_Latency_and_Operation_Mode.png
Figure 8 (mechanical conditions)	Drawn from T06_Comparable_Mechanical_Condition_Efficiency.csv
Table 7 (network-quality groups)	T02_Network_Quality_vs_Efficiency.csv; network-quality group summary cell
Table 8 (latency deciles)	T03_Latency_Decile_Performance.csv; efficiency-by-latency-decile cell
Table 9 (packet-loss deciles)	T04_Packet_Loss_Reliability.csv; packet-loss decile defect cell
Table 10 (operation modes)	T05_Operation_Mode_Summary.csv
Table 11 (mechanical conditions)	T06_Comparable_Mechanical_Condition_Efficiency.csv
Table 12 (final metrics)	T01_Final_Source_of_Truth_Metrics.csv
Table 13 (risk-zone assessment)	Synthesized from T03, T04 and the decile/sensitivity-zone outputs
Figure 1; Tables 1–6, 13, 14	Workflow diagram, literature matrix, and summaries compiled from the notebook audit, descriptive-statistics cells, and Section 4 results